

WINNER ENGINEERING LEARNING HUB SERIES

INTERMEDIATE TOPIC
CAV/VAV & VENTURI
CONTROL SYSTEM



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永胜工程私人有限公司
Excellence in Service

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- ▶ Introduction
- ▶ Sizing of VAV Box & Venturi Valve/Damper
- ▶ Design Considerations & Things to note

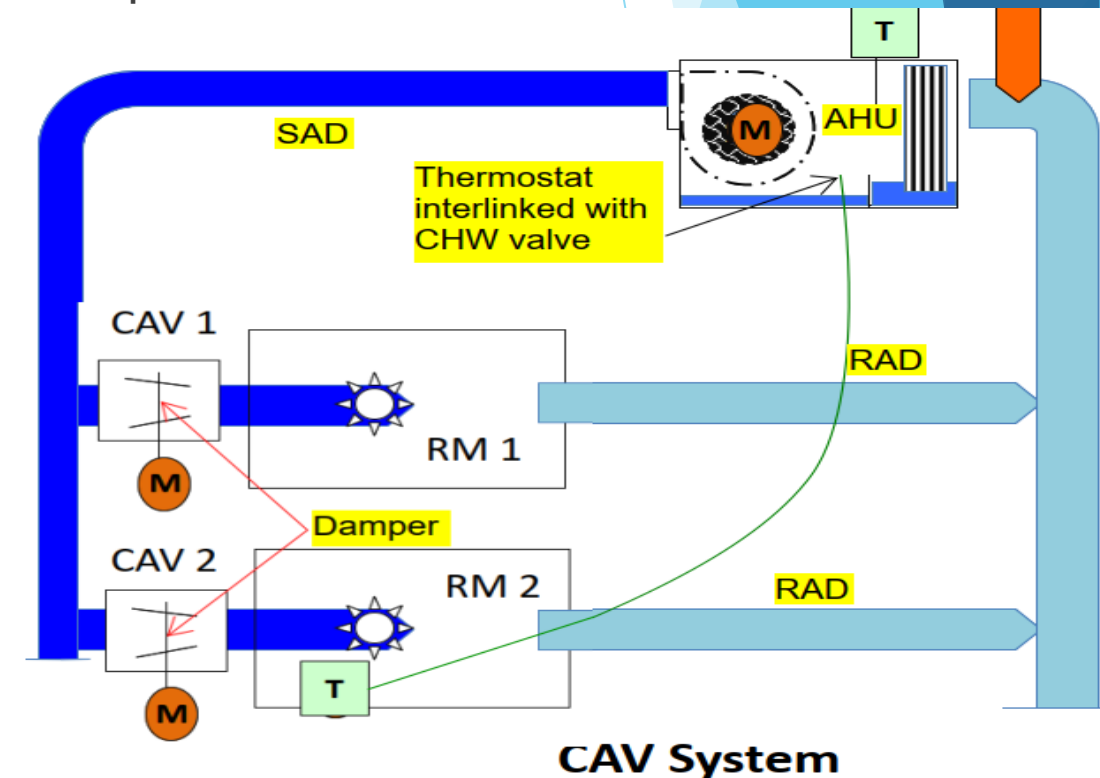
Introduction

Constant Air Volume(CAV) system is designed to...

- ▶ Supply constant airflow to spaces, regardless of the thermal load
- ▶ **Temperature Control** by adjusting the Off-Coil Temperature in AHU
- ▶ Simple and low initial cost system
- ▶ CAV system can be used for pressure regime control (Special Case)

However...

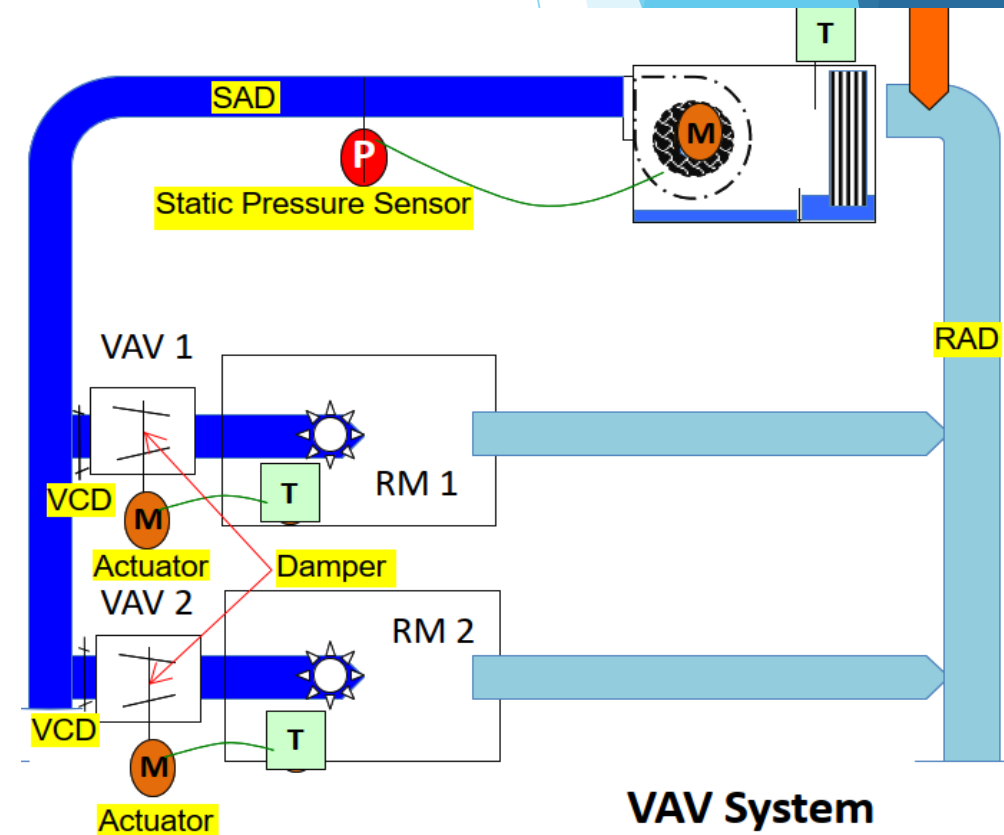
- ▶ Not suitable for spaces with unstable heat loads or with specific temperature/RH control requirements
 - Slow response to load changes
 - Limited temperature control methods
- ▶ Not suitable for multiple spaces with varying heat loads
 - No precise airflow control for individual spaces



Introduction

Variable Air Volume(VAV) system is designed to...

- ▶ Vary the airflow to individual spaces with different heat loads
- ▶ **Temperature Control** by adjusting the amount of airflow through the VAV box & controller in each space
- ▶ Suitable for multiple spaces with variable heat loads
 1. Fast response to heat load changes
 2. Precise Airflow control for individual spaces



Introduction

Types of Variable Air Volume(VAV) Box...

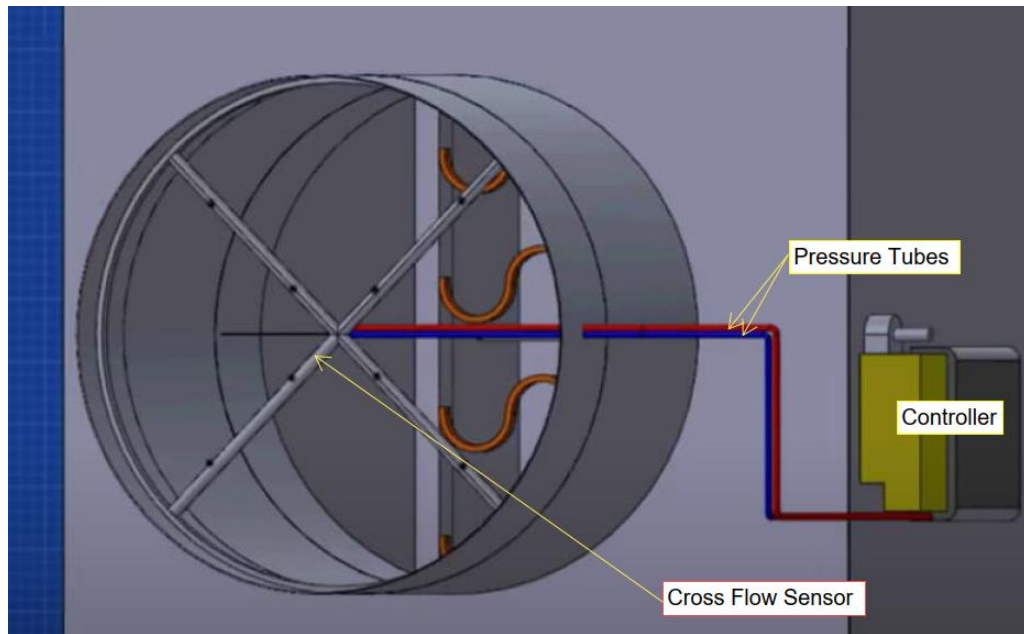
Pressure Dependent(PD)

VS

Pressure Independent(PI)

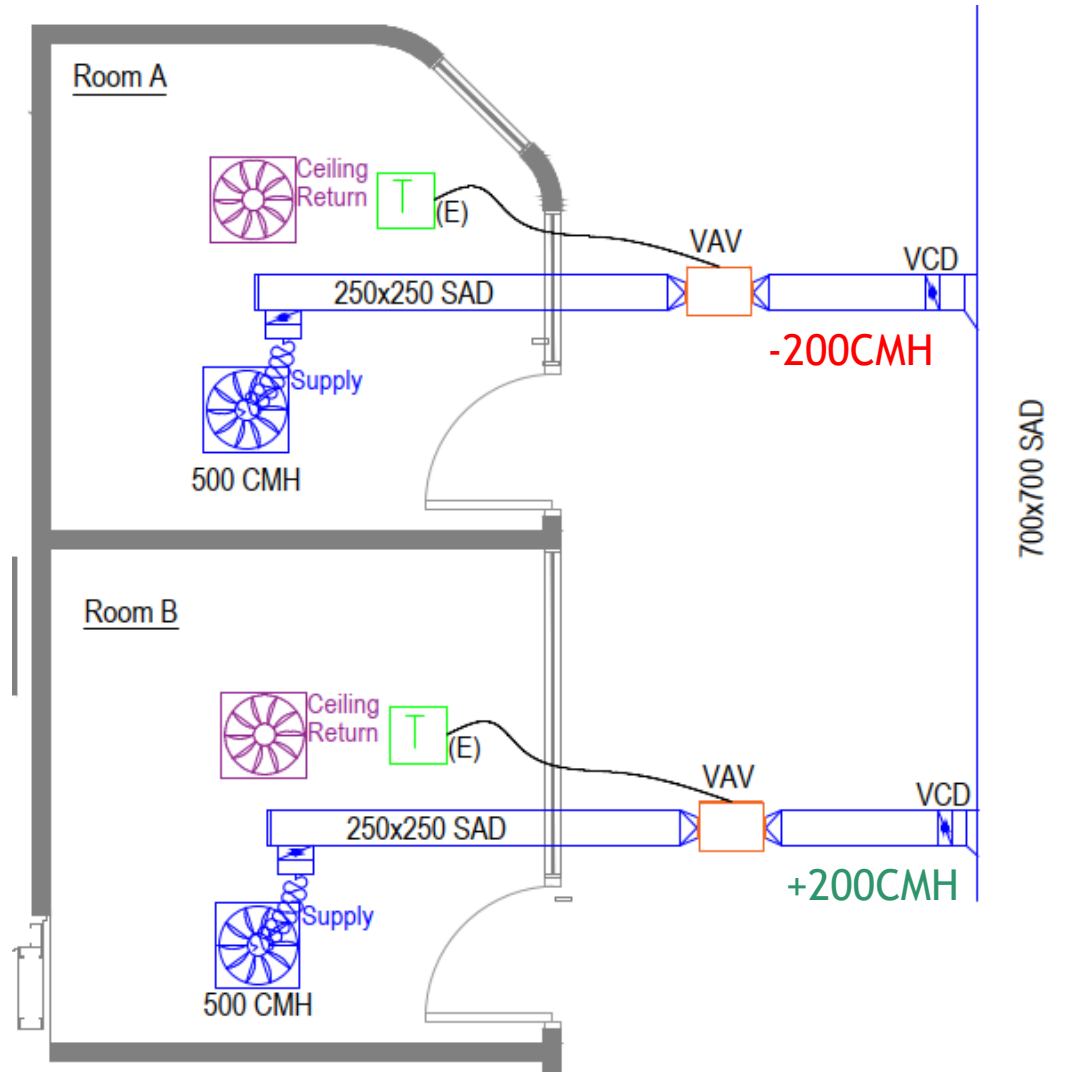
- Without **Cross Flow Sensor**
- Airflow affected by pressure changes in duct (Inconsistent)
- Unstable Room Comfort
- Cheaper

- With **Cross Flow Sensor**
- Airflow not affected by pressure changes in duct (Consistent)
- Stable Room Comfort
- More Expensive



***[Right Click here to find out more on Cross Flow Sensor](#)

Types of Variable Air Volume(VAV) Box (EXAMPLE)...



Introduction

Venturi Damper & Venturi Valves are...

- ▶ Equipments to **maintain a certain pressure regime**
- ▶ Both have tapered section that reduces in size, creating a constriction (mechanism is different)

Venturi Damper

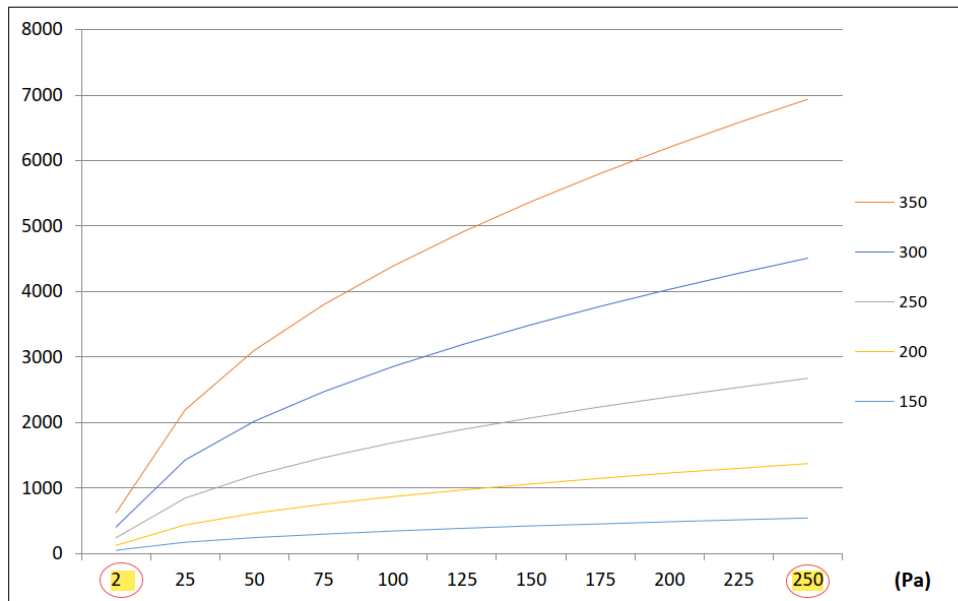
VS

Venturi Valve

- ▶ Pressure drops ranges from 2-250Pa

- ▶ Pressure Drops ranges from 150-750Pa

Flow (CMH)



Differential pressure across venturi damper

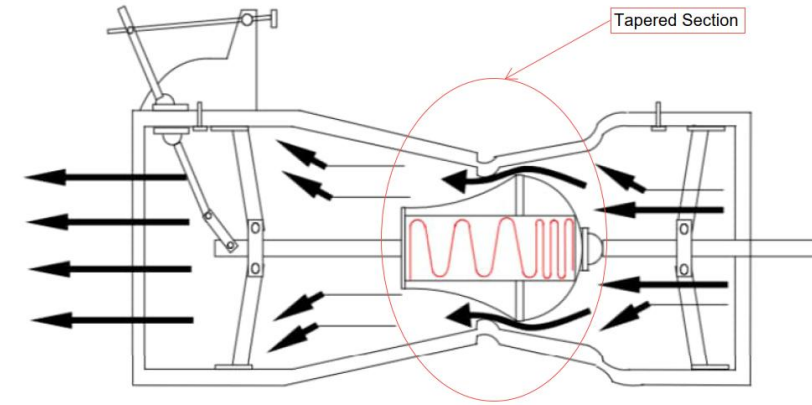
Dynamic III Venturi Valves Features

TRULY PRESSURE INDEPENDENT

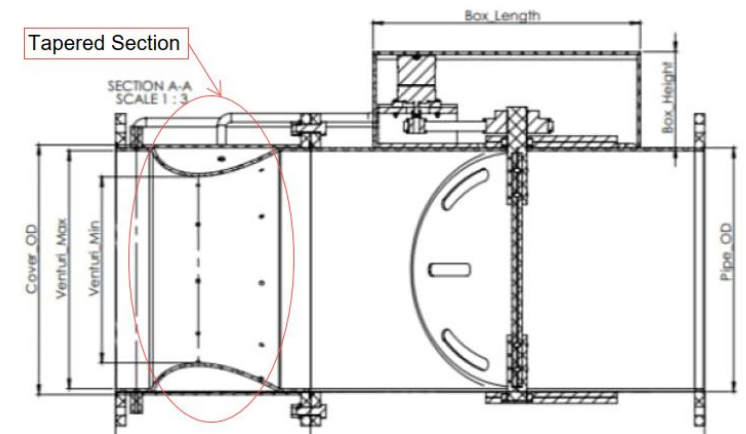
Dynamic III venturi valve is a truly pressure independent airflow using a combination of a calibrated shaft and cone to produce a pressure independent device.

Dynamic III is different from the Traditional Variable Air Volume Box available in the market as the flow controls is separated from the traditional velocity pressure sensing utilized by these commercial devices. The pressure independent control portion is control via the spring loaded cone that compensates automatically when duct static fluctuates between the pressure ranges of **150 PA to 750PA**. The

Venturi Valve (VV)



Venturi Damper (VD)



Introduction

Comparison Summary Page...

	VAV BOX	VENTURI DAMPER	VENTURI VALVE
Airflow Measurement Accuracy	±5% (2.5m/s-12m/s)	±5% (1m/s-12m/s)	±5%
Speed of response	*Fast response VAV Box (1-5s)	Almost Instantaneous (≤1-2s)	Almost Instantaneous (≤1s)
Pressure Independent Control	Yes	Yes	Yes
Control Turndown ratio	6:1	10:1	16:1
Low Pressure Drop @ Operating Range	Yes (2-60Pa)	Yes (2-80Pa)	No (Min. 150Pa)
Corrosion Resistant	Yes (Stainless Steel)	Yes (Stainless Steel)	Yes (Stainless Steel)
Straight Duct Length Requirement for Installation	Yes (min. 1-3xØDia)	No	No

Sizing

Variable Air Volume(VAV) Box Sizing...

- STEP 1: Determine the Minimum & Maximum Airflow (For BMS Configuration)
 - Min. & Max. Airflow setpoints act as extreme limits for the VAV to fluctuate according to the varying heat load.
 - Rule of thumb to be $\pm 20\text{-}30\%$ of the Design Airflow or follow specified requirements

- STEP 2: Select size base on Suppliers' Catalogue

Rule of thumb

- Selection should be based on **Design Airflow**
- Pressure drop to be **10-25Pa**
- Noise Criteria (NC) **<40**

Unit Size	Airflow		Minimum Press. Loss (Pa)	Example: 500CMH						
	CMH	L/S		@125 Pa (0.5"W.G.)						
				Octave Band Centre Frequency						
				125	250	500	1k	2k	4k	NC
5	155	43	+	44	32	33	35	29	20	23
	200	56	+	45	34	35	37	31	22	25
	250	69	11	48	37	38	38	33	24	26
	300	83	17	50	40	40	40	34	26	28
	350	97	24	51	43	43	42	36	27	30
	400	111	32	53	45	45	44	38	29	32
	450	125	40	54	47	47	46	41	33	34
	500	139	48	54	47	49	48	43	37	36
	530	147	55	58	52	51	51	46	39	40
6	250	69	+	41	35	34	35	28	20	23
	300	83	+	43	37	36	37	30	23	25
	350	97	+	44	39	38	39	31	25	27
	400	111	12	45	41	40	41	33	26	29
	450	125	15	45	43	42	42	34	27	31
	500	139	19	46	45	43	43	35	28	32
	550	153	24	46	47	44	45	37	29	33
	600	167	29	46	48	45	46	39	31	35
	650	181	33	47	48	46	47	40	33	36
	700	194	39	47	48	48	49	42	35	37
	750	208	44	47	49	49	50	44	37	38
7	350	97	+	43	41	38	37	31	23	25
	450	125	+	44	43	40	39	33	26	26
	550	153	10	45	46	42	41	35	27	29
	650	181	14	47	47	44	43	37	30	31
	750	208	19	47	49	47	45	40	33	34
	850	236	24	49	51	49	47	42	35	37
	950	264	32	50	53	51	49	44	37	40
	1000	278	35	51	54	52	50	45	38	41

Example:
500CMH

Sizing

Venturi Valve/Damper Sizing...

- Select size base on Suppliers' Catalogue
 - Rule of thumb: Size base on Operating Range

Operating Range

- 10 to 90% non-considering RH Performance
- No additional straight duct runs needed before or after valve
- Available in flow from approx. 35 to 4386 CMH (Ø150mm to Ø350mm)

Size (mm)	Min (CMH)	Max (CMH)
Ø150	35	342
Ø200	87	868
Ø250	170	1691
Ø300	286	2852
Ø350	439	4386

Source: Acesian Ecoflow Venturi Damper

Valve Operating Range

Valve Flow Ranges

Valve Size	Minimum Flow	Maximum Flow	Minimum Pressure Drop across valve
8	35cfm (60 m3/h)	600cfm (1020 m3/h)	0.6 In Wg (150Pa)
10	50cfm (85 m3/h)	850cfm (1445 m3/h)	0.6 In Wg (150Pa)
12	90cfm (153 m3/h)	1300cfm (2210 m3/h)	0.6 In Wg (150Pa)
2-10	100cfm (170 m3/h)	1700cfm (2890 m3/h)	0.6 In Wg (150Pa)
2-12	180cfm (306 m3/h)	2600cfm (4420 m3/h)	0.6 In Wg (150Pa)

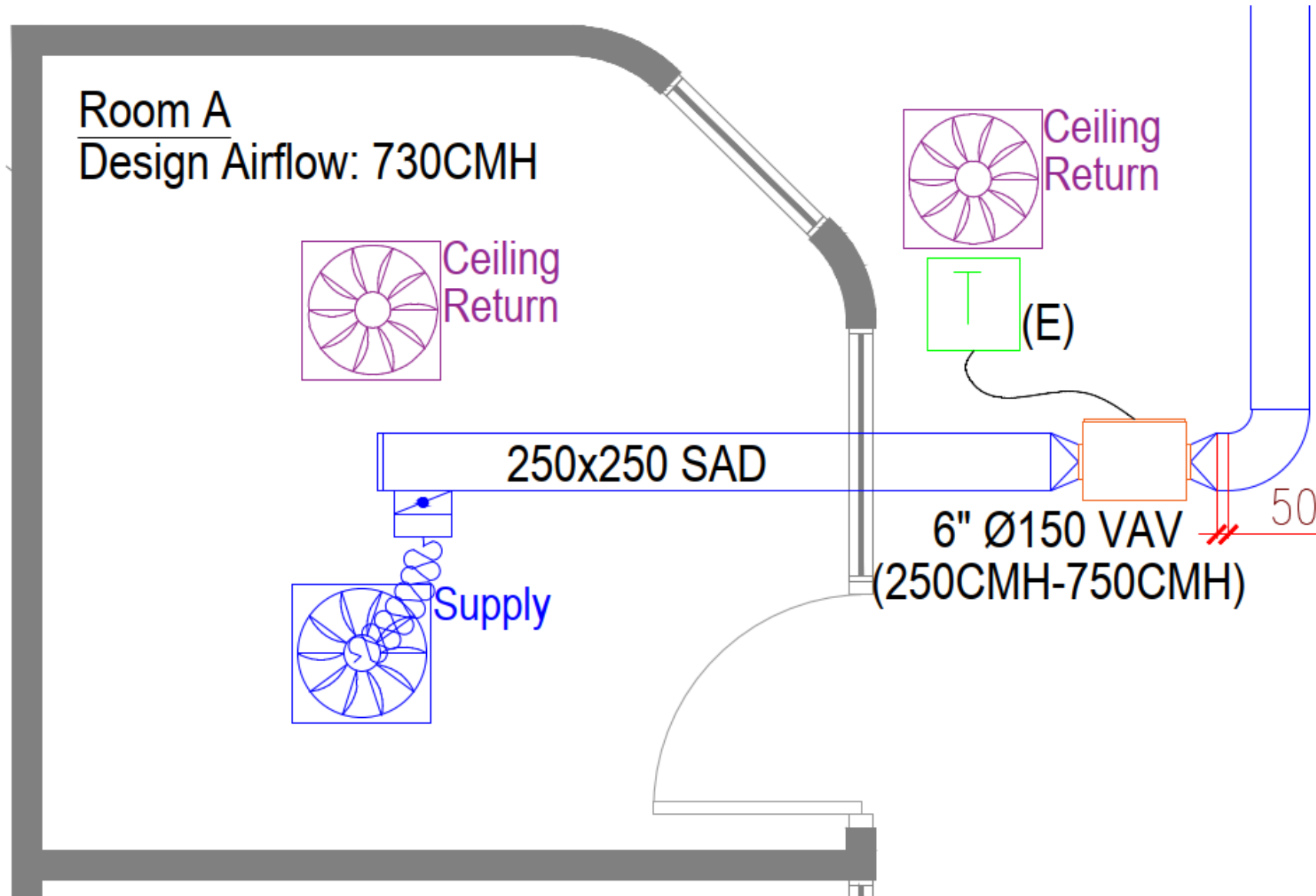
*Note

For flow volume that exceed 4420m3/h please select combination of single and dual valve to meet design volume.

Source: SACO Venturi Valve

Design Considerations & Things to note

Spot the 4 Mistakes



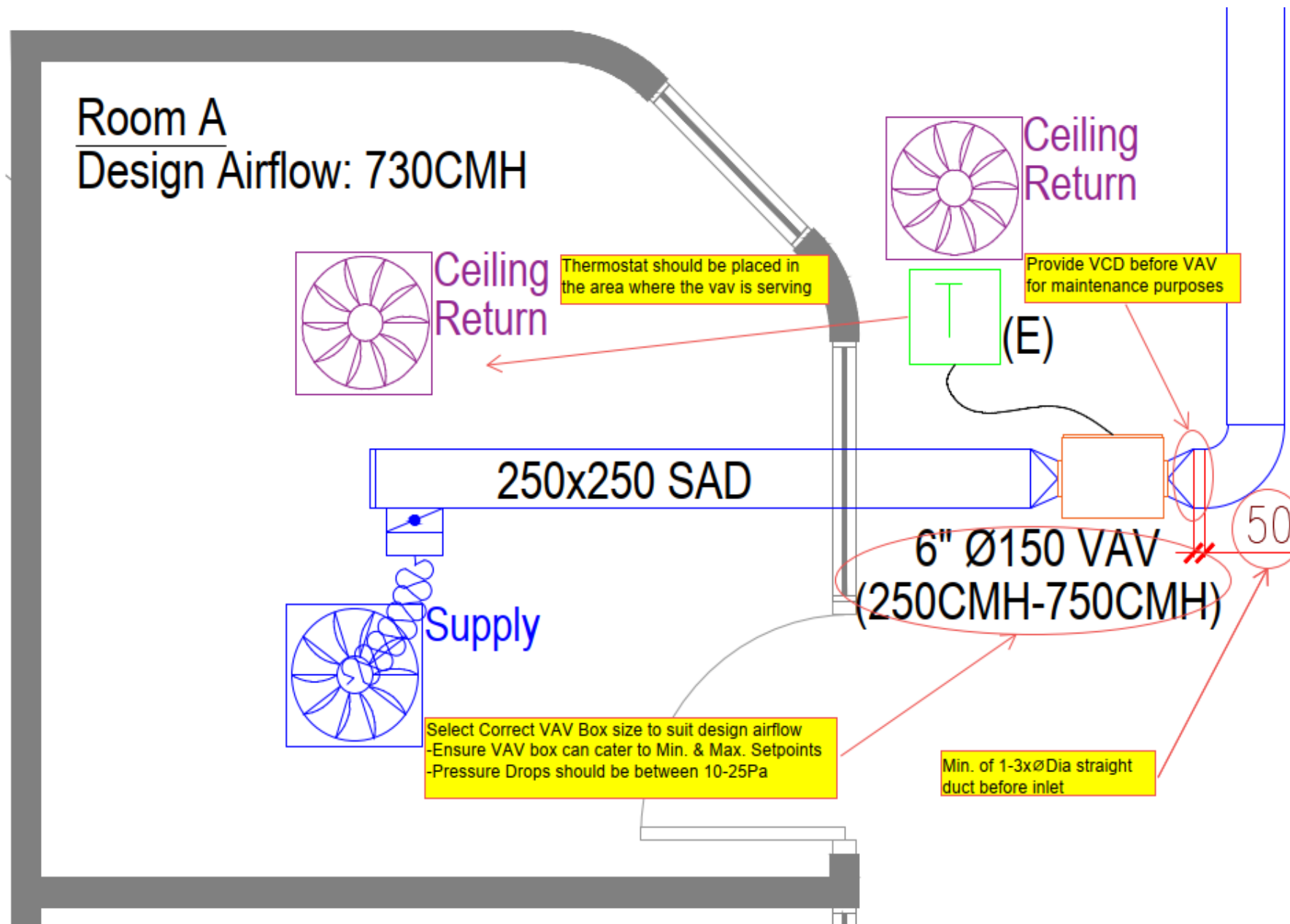
Unit Size	Airflow		Minimum Press. Loss (Pa)
	CMH	L/S	
5	155	43	+
	200	56	+
	250	69	11
	300	83	17
	350	97	24
	400	111	32
	450	125	40
	500	139	48
6	530	147	55
	250	69	+
	300	83	+
	350	97	+
	400	111	12
	450	125	15
	500	139	19
	550	153	24
7	600	167	29
	650	181	33
	700	194	39
	750	208	44
	350	97	+
	450	125	+
	550	153	10
	650	181	14
7	750	208	19
	850	236	24
	950	264	32
	1000	278	35

730

50

VAV Sizing Reference

Design Considerations & Things to note

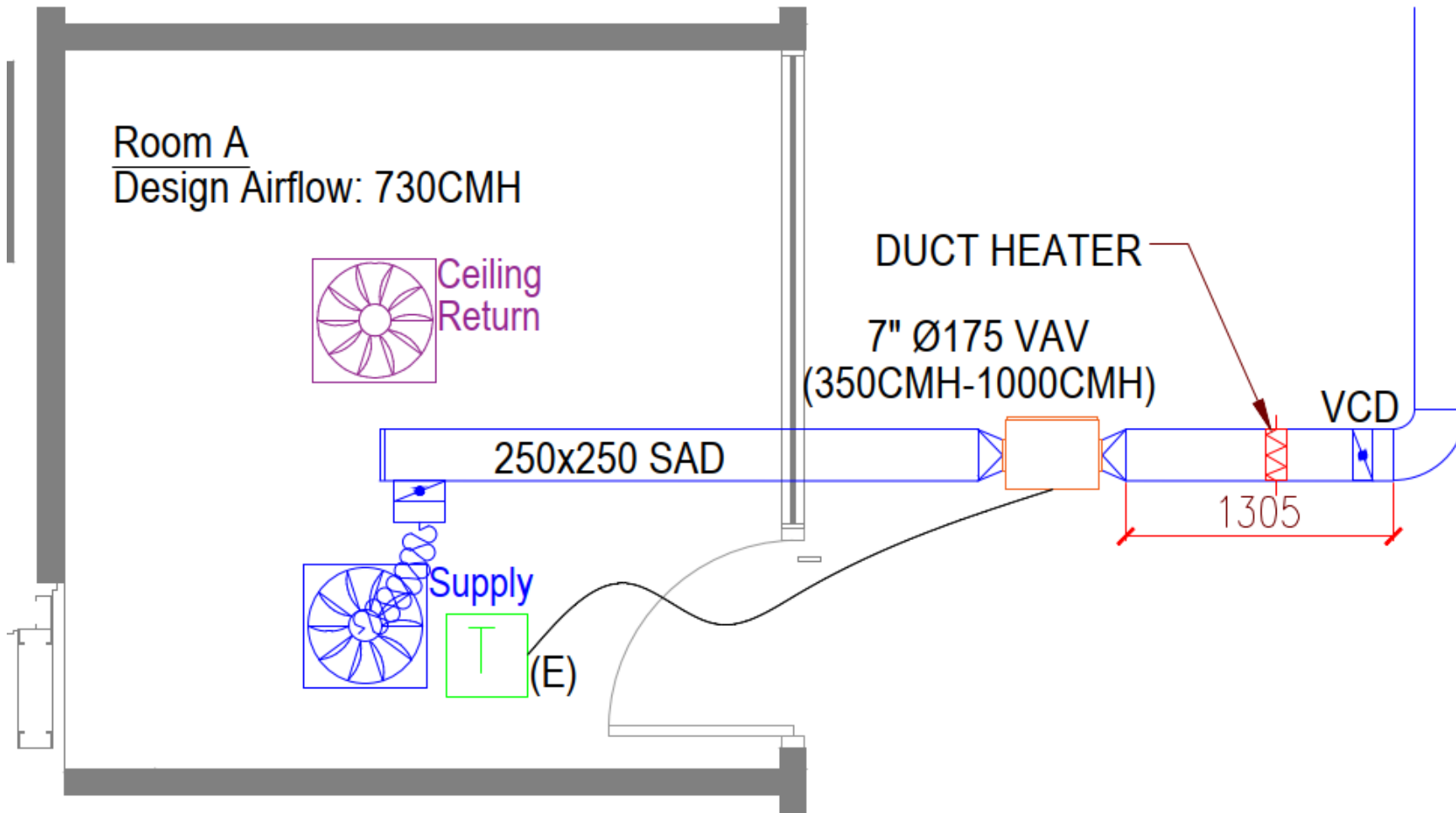


Unit Size	Airflow		Minimum Press. Loss (Pa)
	CMH	L/S	
5	155	43	+
	200	56	+
	250	69	11
	300	83	17
	350	97	24
	400	111	32
	450	125	40
	500	139	48
6	530	147	55
	250	69	+
	300	83	+
	350	97	+
	400	111	12
	450	125	15
	500	139	19
	550	153	24
7	600	167	29
	650	181	33
	700	194	39
	750	208	44
	350	97	+
	450	125	+
	550	153	10
	650	181	14
7	750	208	19
	850	236	24
	950	264	32
	1000	278	35

VAV Sizing Reference

Design Considerations & Things to note

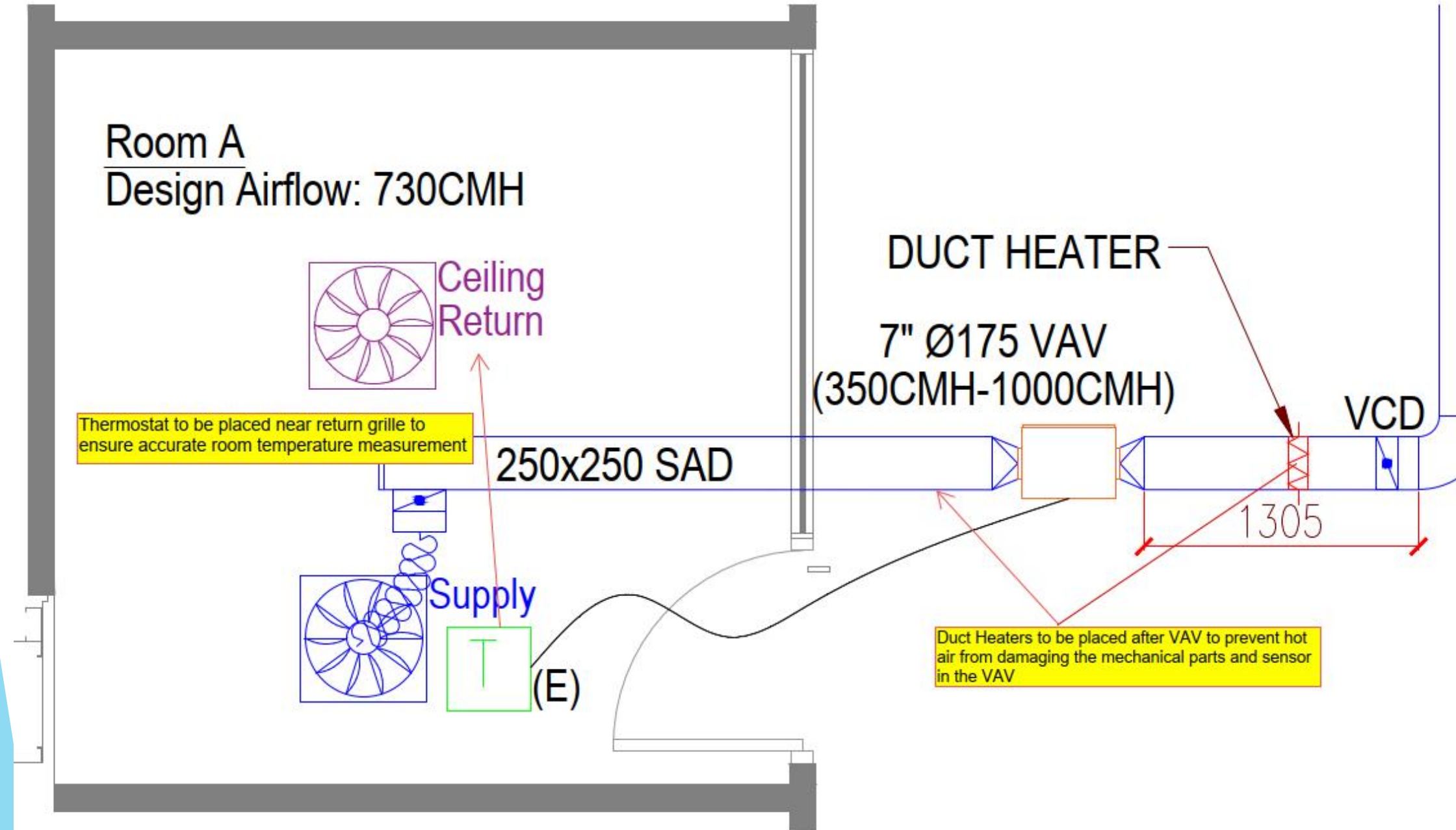
Spot the **2** Mistakes



Design Considerations & Things to note

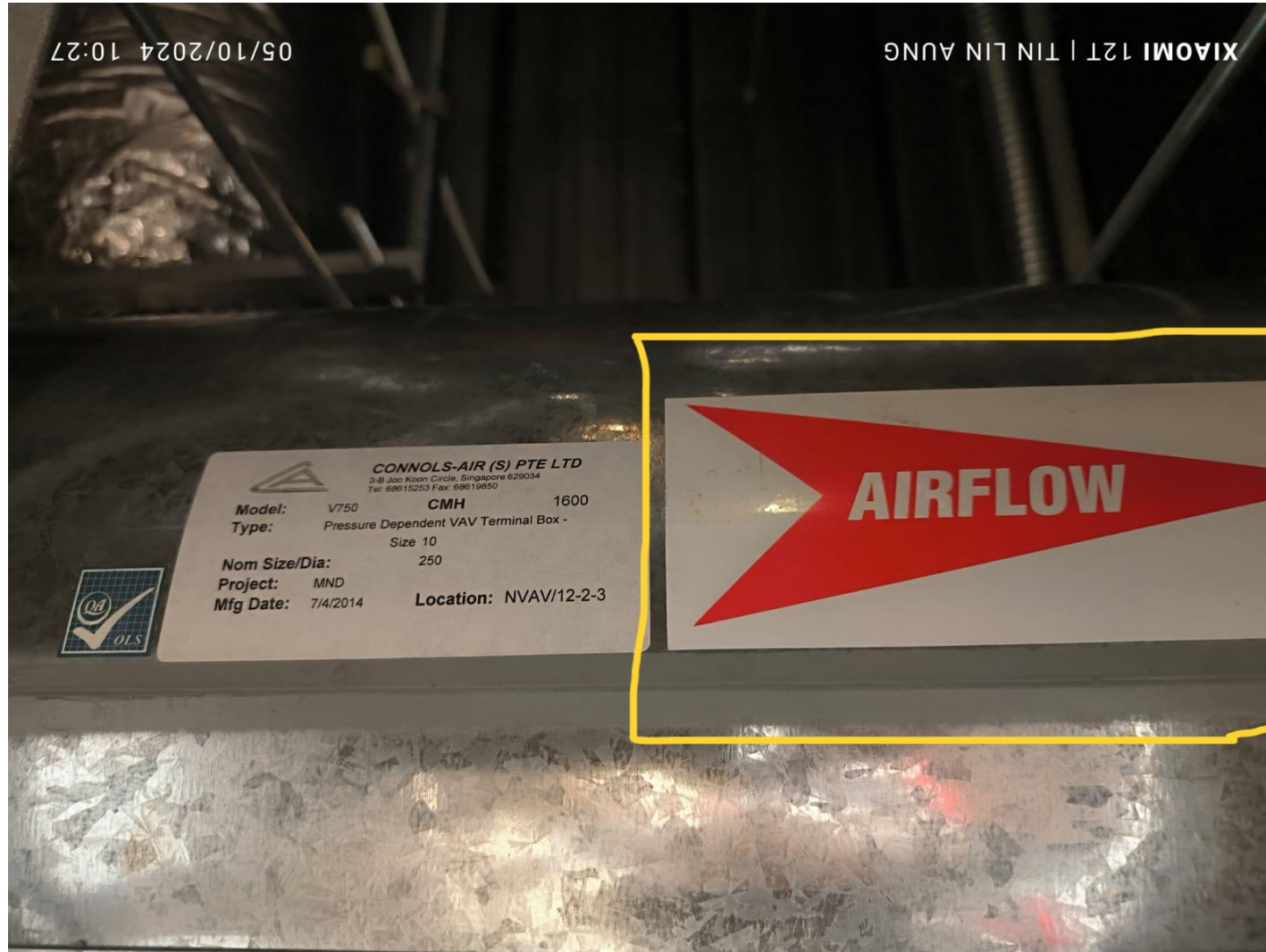
Room A

Design Airflow: 730CMH



Design Considerations & Things to note

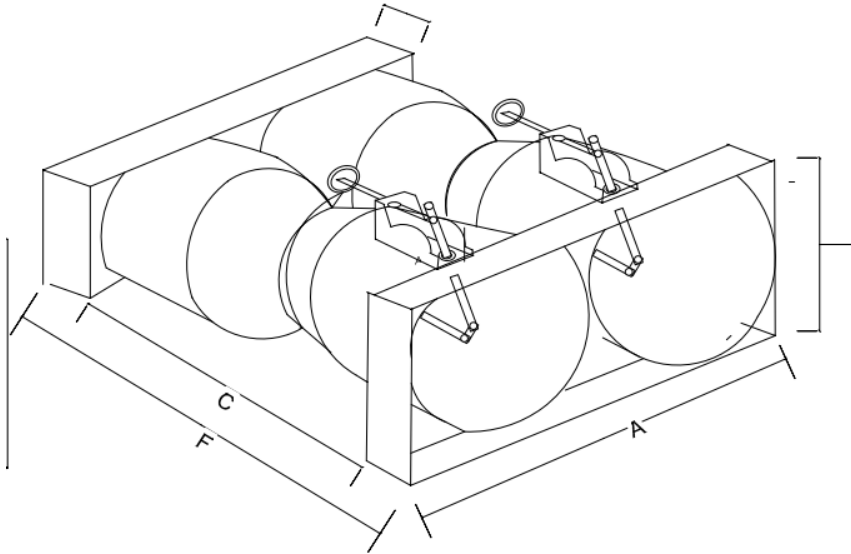
- Make sure Equipment's site installation is in the correct orientation



On site VAV Reference

Design Considerations & Things to note

- Can VAV be connected in parallel manner similarly to Venturi Valve?



Valve Operating Range

Valve Flow Ranges

Valve Size	Minimum Flow	Maximum Flow	Minimum Pressure Drop across valve
8	35cfm (60 m3/h)	600cfm (1020 m3/h)	0.6 In Wg (150Pa)
10	50cfm (85 m3/h)	850cfm (1445 m3/h)	0.6 In Wg (150Pa)
12	90cfm (153 m3/h)	1300cfm (2210 m3/h)	0.6 In Wg (150Pa)
2-10	100cfm (170 m3/h)	1700cfm (2890 m3/h)	0.6 In Wg (150Pa)
2-12	180cfm (306 m3/h)	2600cfm (4420 m3/h)	0.6 In Wg (150Pa)

SACO Dual 10/12 inch Venturi Valve

- Not recommended as it may lead to **turbulent air** which will cause **noise issues**